Semester 1, 2026

Business Plan and Engineering Design Progress Report

EGH419: Advanced Design and Entrepreneurship

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1 Product Name

The proposed product developed by Group 40 is titled **PowerStake**. PowerStake is a smart power management system designed to help households reduce electricity costs through appliance-level monitoring, automated outlet control, predictive insights, and a user-friendly mobile dashboard. The concept combines affordable hardware with intelligent software to reduce standby energy waste, improve convenience, and identify potential appliance faults before failure occurs.

The Group 40 project team is structured using functional roles to allocate responsibilities across business, technical, and design tasks. Team members and primary subsystem responsibilities are listed below:

- **Handler - Erin Keur:** Project coordination and power usage reduction strategy.
- **Hustler - Ethan Bailey:** Commercial strategy, customer value proposition, and mobile dashboard development.
- **Hacker - Andrew Pollard:** Hardware development including power measurement and supply systems.
- **Hipster - Xanthea Vazey:** User experience, industrial design, and automatic power-off scheduling functions.
- **Helper - Jeremy Khin Zaw:** Cross-functional support across all project areas, with primary responsibility for fault detection systems.

This team structure enables parallel development of the core PowerStake subsystems while maintaining shared ownership of prototype integration and validation activities.

2 Target TRL

The current readiness level of the Automatic Power-off Scheduling subsystem is estimated at TRL 2. At present, the concept has been defined and supported through preliminary research into existing technologies such as smart plugs, programmable timers, relay modules, and mobile-controlled outlet systems. However, no integrated prototype has yet been constructed or experimentally validated by the project team.

Given the remaining project schedule, the realistic target for this assessment is TRL 3 progressing toward early TRL 4. This would involve developing and demonstrating a proof-of-concept prototype using commercially available microcontrollers (e.g. ESP32), relay modules, and basic power sensing inputs. Initial validation may be performed on low-voltage test circuits or simulated appliance loads rather than direct 240 V mains systems, reducing safety risk and development complexity.

A successful prototype outcome would demonstrate the core subsystem functions of detecting inactive loads, applying simple shutoff logic, manually restoring power, and communicating with a mobile dashboard interface. Supporting work may also include enclosure concepts developed through CAD modelling and early physical packaging studies for future iterations.

Achieving a full TRL 4 mains-ready prototype within the current timeframe is considered ambitious due to hardware procurement, electrical safety constraints, and software integration requirements. A more practical near-term pathway would be completion of a validated bench-scale prototype during this unit, followed by future progression to TRL 4-5 through controlled

household testing across common appliances such as televisions, chargers, gaming consoles, and kitchen devices.

3 Problem Validation and Engineering Design

3.1 Customer Needs

Many households waste electricity through standby loads from televisions, chargers, gaming consoles, printers and kitchen appliances [1]. While individual standby loads are small, cumulative yearly costs are meaningful. Existing timer switches require manual programming and do not adapt to user behaviour. Most smart plugs offer remote switching but still depend on active user involvement. Customers require a solution that automatically reduces unnecessary consumption while maintaining convenience. The system must be simple to use, safe, and allow manual override when required.

3.2 High-Level Objective (HLO-4)

The subsystem shall automatically recommend or execute outlet shutoff schedules that reduce standby energy consumption by at least 40% during controlled testing, while allowing manual user override within 2 seconds.

4 System Requirements

The system requirements for the Automatic Power-off Scheduling subsystem have been developed to ensure alignment with HLO-4 and the broader PowerStake prototype objectives. Requirements are divided into Physical, Functional, and Performance categories. Physical requirements define hardware constraints and design limitations. Functional requirements define what the subsystem must do. Performance requirements define how effectively the subsystem must operate and be validated.

4.1 Numbering Scheme

Each requirement is identified using the following format:

- REQ-Ph-XX = Physical Requirement
- REQ-F-XX = Functional Requirement
- REQ-Pe-XX = Performance Requirement

4.2 Physical Requirements

Table 1: Physical subsystem requirements

ID	Requirement Description	Trace
REQ-Ph-01	The subsystem shall control a minimum of four independently switched AC outlets.	HLO-4
REQ-Ph-02	The subsystem shall operate from Australian 240V mains supply using isolated switching hardware.	HLO-4
REQ-Ph-03	The subsystem shall include wireless communication hardware for dashboard connectivity.	HLO-4
REQ-Ph-04	The prototype enclosure footprint shall remain below 350 mm $\times$ 150 mm.	HLO-4
REQ-Ph-05	The subsystem shall utilise low-voltage internal control circuitry separated from switched mains outputs.	HLO-4

4.3 Functional Requirements

Table 2: Functional subsystem requirements

ID	Requirement Description	Trace
REQ-F-01	The subsystem shall detect outlet power state at intervals no greater than 1 second.	HLO-4
REQ-F-02	The subsystem shall identify standby loads below a configurable wattage threshold.	HLO-4
REQ-F-03	The subsystem shall generate shutoff recommendations using repeated inactivity patterns.	HLO-4
REQ-F-04	The subsystem shall automatically disconnect selected outlets according to approved schedules or inactivity rules.	HLO-4
REQ-F-05	The subsystem shall allow manual override through the mobile dashboard.	HLO-4
REQ-F-06	The subsystem shall restore outlet power upon authorised user command.	HLO-4
REQ-F-07	The subsystem shall log switching actions for later user review.	HLO-4

4.4 Performance Requirements

Table 3: Performance subsystem requirements

ID	Requirement Description	Trace
REQ-Pe-01	The subsystem shall reduce standby energy consumption by at least 40% during controlled prototype testing.	HLO-4
REQ-Pe-02	Automatic shutoff commands shall execute within 2 seconds of trigger conditions being met.	HLO-4
REQ-Pe-03	At least 80% of schedule recommendations shall be accepted during five representative user test scenarios.	HLO-4
REQ-Pe-04	Wireless command success rate shall exceed 95% during controlled testing.	HLO-4
REQ-Pe-05	The subsystem shall operate continuously for 24 hours without requiring reset during bench testing.	HLO-4

4.5 Verification Matrix

Table 4: Requirement traceability matrix

Requirement ID	HLO-4
REQ-Ph-01	X
REQ-Ph-02	X
REQ-Ph-03	X
REQ-F-01	X
REQ-F-02	X
REQ-F-03	X
REQ-F-04	X
REQ-F-05	X
REQ-F-06	X
REQ-Pe-01	X
REQ-Pe-02	X
REQ-Pe-03	X
REQ-Pe-04	X
REQ-Pe-05	X

4.6 Subsystem Architecture

The AI subsystem forms the analytical layer of the PowerStake platform. It receives historical and near real-time power monitoring data from the onboard circuitry through the Wi-Fi and cloud communication path. Its role is to transform raw electrical measurements into useful outputs such as appliance fault alerts, power-saving opportunities, and recommended automatic shutoff schedules.

The subsystem contains four main stages: data ingestion, preprocessing, inference, and output generation. During data ingestion, measured voltage, current, power, and switching history

are received and stored. These data are then filtered, normalised, and converted into useful features such as standby duration, repeated inactivity, and unusual load changes. The inference stage applies analytical logic to classify appliance behaviour as normal, inefficient, or potentially faulty. Finally, the output stage sends recommendations and alerts to the mobile dashboard, where users can review insights and approve or reject suggested actions.

This subsystem interfaces directly with the wider PowerStake monitoring, communication, and dashboard environment, and provides the decision-support capability required for intelligent energy management.

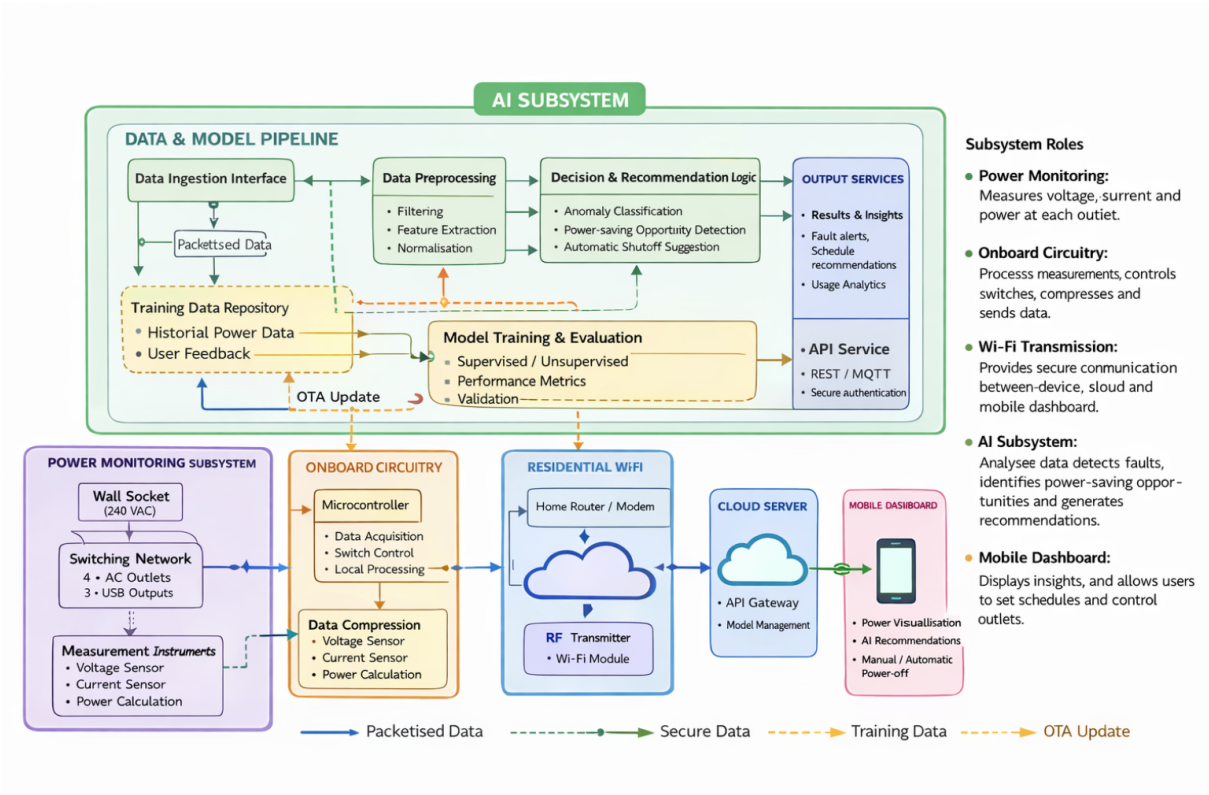

Figure 1: Subsystem architecture for the PowerStake AI subsystem [2].

4.7 Trade Studies

A trade study was conducted to determine the most suitable analytical approach for the PowerStake AI subsystem. The subsystem is required to analyse power monitoring data to identify appliance malfunctions, detect power-saving opportunities, and recommend automatic shutoff actions. For the current prototype stage, the preferred solution needed to be accurate enough to demonstrate value, while remaining practical to implement with limited data, modest computational resources, and a short development timeframe.

Three candidate approaches were evaluated: (1) rule-based thresholding, (2) classical machine learning, and (3) lightweight deep learning. These options were compared based on fault detection capability, data requirement, computational demand, interpretability, and prototype suitability.

Option	Description	Data Requirement	Computational Demand	Suitability
Rule-Based Thresholding	Fixed rules using power, current, and inactivity duration	Low	Low	High
Classical Machine Learning	Feature-based classification using engineered inputs	Medium	Medium	Medium
Lightweight Deep Learning	Learns temporal patterns directly from power traces	High	High	Low

Table 5: Initial comparison of AI modelling approaches

Option	Detection	Data Need	Processing	Interpretability	Total
Rule-Based Thresholding	3	5	5	5	18
Classical Machine Learning	4	3	3	3	13
Lightweight Deep Learning	5	2	2	2	11

Table 6: Comparison matrix for AI subsystem options

Rule-based thresholding was selected as the preferred prototype solution. Although it is less sophisticated than machine learning methods, it is the most realistic option for the current project stage because it requires minimal training data, is easy to interpret, and can be implemented rapidly. Classical machine learning is a strong future option once more operating data become available. Lightweight deep learning was considered less suitable at this stage due to higher computational and data requirements. Overall, rule-based thresholding provides the best balance of practicality, transparency, and prototype suitability for demonstrating the AI subsystem value proposition.

References

[1] U. D. of Energy. "Standby power and energy vampires." Explains standby power consumption from household electronics and methods to reduce wasted energy, Accessed: Apr. 17, 2026. [Online]. Available: `https://www.energy.gov/energysaver/standby-power`.

[2] OpenAI, *System architecture figure generated using chatgpt*, AI-generated image created for report use on 17 April 2026, 2026.